PW 2

1 Project Creation & Setup

Just like last **PW** just create an empty views activity project

2 Application

In this **PW** we will create an application that takes as input a number and computes its double

2.1 Colors.xml

We will create a new color in `res/values/colors.xml` that we will use later in the layout

colors.xml

```xml
<?xml version="1.0" encoding="utf-8"?>
<resources>
    <color name="black">#FF000000</color>
    <color name="white">#FFFFFFFF</color>
    <color name="result">#F32FA23B</color> <!-- new color i added -->
</resources>
```

2.2 Layout

We're gonna modify the existing layout `res/layout/activity_main.xml` and to do so we will use :

- **LinearLayoutCompat:** A layout that makes each component appear either one below the other if `orientation` is set to `vertical`, or one right after the other if set to `horizontal`.
- **TextView:** A component used for text labels.
- **EditText:** A text field component.
- **Button:** A button component.
- **IDs:** Unique identifiers that can be fetched in the context.
- **SP & DP Units:** Independent responsive units **SP** for text size and **DP** for component size and spacing.

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<!-- we set the orientation to vertical, set the context to MainActivity and layout to LinearLayoutCompat
that takes entire screen (math_parent) and alignment to center (gravity) -->
<androidx.appcompat.widget.LinearLayoutCompat xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical"
    tools:context=".MainActivity">

    <!-- "hello" bold label with 25sp text size that takes the minimum space (wrap_content)
    and has margin bottom of 20dp -->
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="hello"
        android:textSize="25sp"
        android:textStyle="bold"
        android:layout_marginBottom="20dp"
        />

    <!-- textfield that only takes decimal number with an id "field"
    takes the entire screen width(match_parent) but minimum height (wrap_content)
    margin left right of 12dp (horizontal)
    margin bottom of 20dp
    -->
    <EditText
        android:id="@+id/field"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginHorizontal="12dp"
        android:hint="Input Number"
        android:inputType="numberDecimal"
        android:layout_marginBottom="20dp"
        />

    <!-- "compute "button with an id "compute_button"
    takes the entire screen width(match_parent) but minimum height (wrap_content)
    margin left right of 40dp (horizontal)
    margin bottom of 20dp
    when clicked it will call the method "computeDouble" in the context "MainActivity"
    -->
    <Button
        android:id="@+id/compute_button"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginHorizontal="40dp"
        android:text="compute"
        android:onClick="computeDouble"
        android:layout_marginBottom="20dp"
        />

    <!-- "result : " label with 15sp text size that takes the minimum space (wrap_content)
    with an id "result_label"
    color of result that we defined in colors.xml
    -->
    <TextView
        android:id="@+id/result_label"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="result : "
        android:textSize="15sp"
        android:textColor="@color/result"
        />

</androidx.appcompat.widget.LinearLayoutCompat>
```

2.3 Context

We need to implement the `computeDouble` but before that we need to reference the component of the layout in the context to do so we will use the `findViewById` which fetches the component by id (same id defined in layout) and we will create this link in `onCreate` method

MainActivity.java

```java
package com.example.tp2;

import static android.content.ContentValues.TAG;
import android.os.Bundle;
import android.util.Log;
import android.view.View;
import android.widget.EditText;
import android.widget.TextView;
import android.widget.Toast;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {

    // defining the components
    private EditText numberField;
    private TextView result;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);

        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);

        // fetching the components by id
        result = findViewById(R.id.result_label);
        numberField = findViewById(R.id.field);

        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) -> {
            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
            return insets;
        });
    }


    public void computeDouble(View view) {
        // we fetch the text in field and trim it
        String valueField = numberField.getText().toString().trim();

        if (valueField.isEmpty()) {
            // if empty create a toast to notify user that field is empty and exit the method
            Toast.makeText(this, "please input a number", Toast.LENGTH_LONG).show();
            return;
        };

        // else parse text to decimal , compute its double and update the result label
        double number = Double.parseDouble(valueField);
        number *=2;
        result.setText("result : "+number);

    }
}
```

3 Execution

Im still executing the application in my physical android phone below are the screenshot of the execution :





